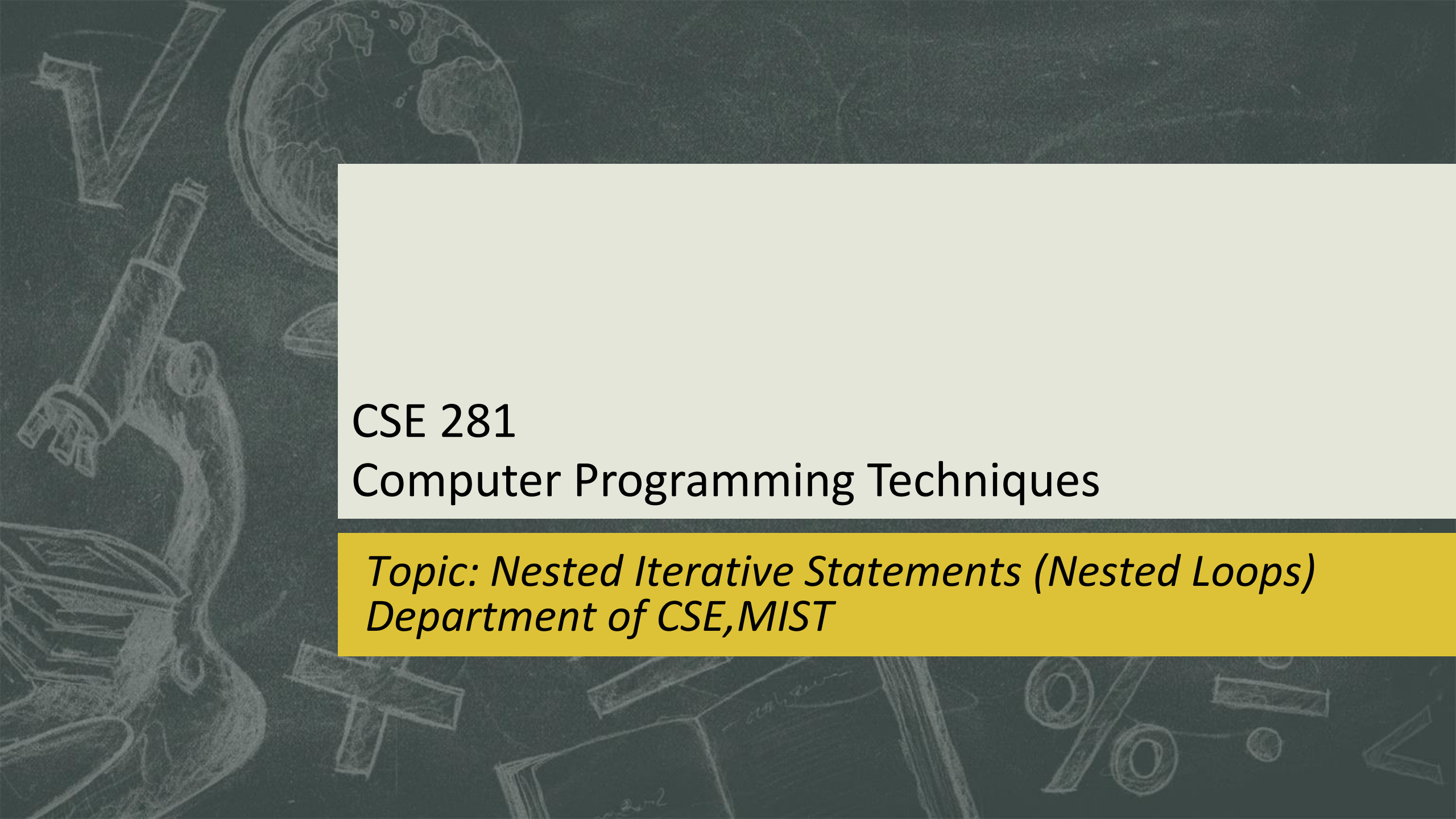




CSE 281

Computer Programming Techniques

Topic: Nested Iterative Statements (Nested Loops)
Department of CSE, MIST

Nested Loop

- Just like nested if else **C** allows the looping of statements inside another **loop**
- There is **no restriction** for defining any number of loops. The **nesting level** can be defined at **n times**.

▪Example :

```
Outer_loop {  
    //outer loop statements...  
    Inner_loop {  
        // inner loop statements.  
    }  
    // outer loop statements.  
}
```

Nested for loop Syntax

```
for(initialization; condition; update loop variable) {  
    // statement 1  
    .....
```

```
for(initialization; condition; update loop variable) {  
    // Inner loop statements  
}  
    // statement n  
}
```

Nested while loop Syntax

```
while(condition)
{
    //outer loop statements.
    while(condition)
    {
        // inner loop statements.
    }
    // outer loop statements.
}
```


Nested do...while loop Syntax

```
do {  
  // statements
```

```
  do {  
    // Inner loop statements  
  }while(condition);
```

```
  // statements
```

```
}while(condition);
```

Nested for loop Syntax

```
for(outVariable initialization ; condition; update outVariable) {  
    // statement 1  
    .....  
    for(inVariable initialization; condition; update inVariable) {  
        // Inner loop statements  
    }  
    // statement n  
}
```

outer
loop

inner
loop

Nested for loop Example

```
for(i=1 ; i<=3 ; i++){  
    printf("*");  
}
```

Sample Output

Nested for loop Example

```
for(i=1 ; i<=3 ; i++){  
    for(j=1 ; j<=3; j++){  
        printf("*");  
    }  
}
```

**Outer loop indicates/defines the row
Inner loop indicated/defines the column**

Desired Output

```
***  
***  
***
```

Output for this code

```
*****
```

How can we solve?

Where should we put the "\n"?

Nested for loop Example

```
for(i=1 ; i<=3 ; i++){  
    for(j=1 ; j<=3; j++){  
        printf("*");  
    }  
    printf("\n");  
}
```

Sample Output

```
***  
***  
***
```

Outer loop indicates/defines the row
Inner loop indicated/defines the column

Nested for loop Example Code

```
for(i=1; i<=2; i++) {  
    for(j=1; j<=2; j++) {  
        printf("i:%d, j:%d\n",i,j);  
    }  
}
```

Outer loop			Inner loop		
Value of i	condition	Execution	Value of j	condition	Execution
1	1<=2, True	Execute inner loop	1	1<=2,true	i:1,j:1
			2	2<=2,true	i:1,j:2
			3	3<=2,false	
2	2<=2,True	Execute inner loop	1	1<=2,true	i:2,j:1
			2	2<=2,true	i:2,j:2
			3	3<=2,false	
3	3<=2,False				

Nested for loop Example (printing pattern)

```
for(i=1 ; i<=n ; i++){  
    for(j=1 ; j<=i; j++){  
        printf("%d",j);  
    }  
    printf("\n");  
}
```

Take input of a number n and print a pattern

Input: n=3

Sample Output

1
12
123

Input: n=4

Sample output

1
12
123
1234

Note: CT 2 Syllabus is up to this.

Nested for loop Example 5(printing pattern)

```
for(i=1 ; i<=n ; i++){  
    for(j=0 ; j<i; j++){  
        printf("%d",n-j);  
    }  
    printf("\n");  
}
```

Take input of a number n and print a pattern

Input: n=3

Sample Output

3

32

321

Input: n=4

Sample output

4

43

432

4321

Nested for loop Example 5 (Alternate Solution)

```
for(i=1 ; i<=n ; i++){  
    int printvalue=n;  
    for(j=0 ; j<i; j++){  
        printf("%d", printvalue);  
        printvalue--;  
    }  
    printf("\n");  
}
```

Take input of a number n and print a pattern

Input: n=3

Sample Output

3

32

321

Input: n=4

Sample output

4

43

432

4321

Nested for loop Example 6

```
for(i=1;i<=n;i++){
```

```
    for(j=1; j<=n-i;j++){  
        printf(" ");  
    }
```

→ For printing space

```
    for(k=1; k<=i ;k++){  
        printf("%d",k);  
    }
```

→ For printing numbers

```
    printf("\n");
```

```
}
```

Take input of a number n and print a pattern

Input: n=3

Sample Output

		1
	1	2
1	2	3

Nested for loop Example 7

```
for(i=1;i<=n;i++){
```

```
    for(j=1; j<=n-i;j++){  
        printf(" ");  
    }
```

→ For printing space

```
    for(k=1; k<=2*i-1 ;k++){  
        printf("*");  
    }
```

→ For printing stars

```
    printf("\n");
```

```
}
```

Take input of a number n and print a Triangle

Input: n=3

Sample Output

		*		
	*	*	*	
*	*	*	*	*

Nested for loop Example 8

```
for(i=1;i<=n;i++){
```

```
    for(j=1; j<=n-i;j++){  
        printf(" ");  
    }
```

→ For printing space

```
    for(k=1; k<=2*i-1 ;k++){  
        printf("%d",k);  
    }
```

→ For printing numbers

```
    printf("\n");
```

```
}
```

Take input of a number n and print a Triangle

Input: n=3

Sample Output

		1		
	1	2	3	
1	2	3	4	5

Nested for loop Example 9

```
for(i=1;i<=n;i++){
```

```
    for(j=1; j<=n-i;j++){  
        printf(" ");  
    }
```

For printing
space

```
    for(k=1; k<=i ;k++){  
        printf("%d",k);  
    }
```

For printing first
triangle

```
    for(l=i ; l>1 ;l--){  
        printf("%d",l-1);  
    }
```

For printing second
triangle

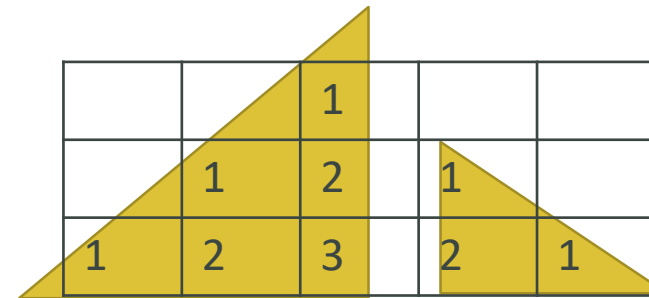
```
    printf("\n");
```

```
}
```

Take input of a number n and print a Triangle

Input: n=3

Sample Output



Nested for loop Sample Problems

For input $n = 3$

1	2	3	2	1
	1	2	1	
		2		

1		
2	3	
4	5	6

1				
1	2	1		
1	2	3	2	1

Thank
you!